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Virtual University	CS401-ASSEMBLY LANGUAGE (SOLVED MCQs) FROM MIDTERM PAPERS LECTURE (1-22)	VU LMS HELP vulmshelp.com
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Question No:1

The first instruction of “COM” file must be at offset:

► 0x0010

► 0x0100 (Page 19)

► 0x1000

► 0x0000

Question No:2

The execution of the instruction “mov word [ES : 0], 0x0741” will print character “A” on screen , background color of the screen will be

► Black (Page 81)

► White

► Red

► Blue

Question No:3

The iAPX888 architecture consists of_____register.

►12

►14

► 16 (Page 15)

►18

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Question No:3

The iAPX888 architecture consists of_____register.

►12

►14

► 16 (Page 15)

►18

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Question No:5

One screen location corresponds to a

► Byte

► Word (Page 80)

► Double byte

➢ Double word

Question No:6

When an item is pushed on the decrementing stack, the top of the stack is

► First decremented and then element copied on to the stack (Page 68)

► First incremented and then element copied on to the stack

► Decremented after the element copied on to the stack

► Incremented after the element copied on to the stack

Question No:7

one Each screen location corresponds to a word, the lower byte of this word contains _____

► The character code (Page 81)

► The attribute byte

► The parameters

► The dimensions

Question No:8

if ax contains decimal -2 and BX contains decimal 2 then after the execution of instructions: CMP AX, BX ,JA label

► Jump will be taken

► Zero flag will set

► ZF will contain value -4

► Jump will not be taken

Question No:9

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If D is “35” is shift to left 2 bits the new value

► 35

►70

► 140

►17

Question No:10

When two 16bit numbers are added the answer can be 17 bits long, this extra bit that won’t fit in the target register is placed in the where it can be used and tested

► carry flag (Page 16)

► Parity Flag

► Auxiliary Carry

► Zero Flag

Question No: 11

Only instructions allow moving data from memory to memory.

► string (Page 29)

► word

► indirect

► stack

Question No:12

When a 16 bit number is divided by an 8 bit number, the quotient will be in

► AL (Page 85)

► AX

► AH

► DX

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Question No:13

one Which bit of the attributes byte represents the red component of background color ?

► 3

► 4

► 5

► 6 (Page 81)

Question No:14

| 0 |--›| 1 | 1 | 0 | 1 | 0 | 0 | 0 | --›| C | is a example of _____

► Shl

► sar

► Shr (Page 52)

► Sal

Question No:15

allow changing specific processor behaviors and are used to play with it.

► Special Instructions (Page 14)

► Data Movement Instructions

► Program Control Instructions

► Arithmetic and Logic Instructions

Question No:16

8088 is a 16bit processor with its accumulator and all registers of_____.

► 32 bits

► 6 bits

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► 16 bits (Page 14)

► 64 bits

Question No:17

decrements SP (the stack pointer) by two and then transfers a word from the source operand to the top of stack

► PUSH (Page 71)

► POP

► CALL

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►RET

Question No: 18

In instruction ADC the operands can be

- Two register only
- Two register and one memory location
- **CF and two other operands (Page 57)**
- ZF and two other operands

Question No:19

After the execution of instruction "RET"

- **SP is incremented by 2 (Page 66)**
- SP is decremented by 2
- SP is incremented by 1
- SP is decremented by 1

Question No: 20

The extended ASCII has

- 64 characters
- **128 characters (Page 79)** AL-JUNAID INSTITUTE OF GROUP
- 256 characters
- 502 characters

Question No:21

The second byte in the word designated for screen location holds

o The dimension of the screen

- Character position on the screen
- **Character color on the screen (Page 81)**
- ACSII code of the character

Question No: 22

REP will always o Incremented CX by 1

- Incremented CX by 2
- **Decremented CX by 1 (Page 92)**
- Decremented CX by 2

Question No:23

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The routine that executes in response to an INT instruction is called

- **ISR (Page 103)**
- IRS
- ISP
- IRT

Question No:24

In the instruction "CMP AX,BX" the contents of

- AX are changed
- BX are changed
- CX are changed
- **Flag register are changed (Page 39)**

Question No:25

All the addressing mechanisms iniAPX88 return a number called_____ address

- **. o Effective (Page 33)**
- faulty
- indirect
- direct

Question No:26

The execution of the instruction "mov word [ES: DI], 0x0720"

- **will clear next character on screen (Page 82)**
  o will print "20" at top left of the screen
- will print "20" at top right of the screen
- will move DI at location 0720 on the screen

Question No: 27

"mov byte [num1],5" is_____instruction.

- **legal (Page 30)**
- illegal
- stack based
- memory indirect

Question No:28

MOV instruction transfers a byte or word from which of the following source location.

- DS:DI

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- ES:SI
- ES:DI
- **DS:SI (Page 92)**

Question No:29

If AX contains FFFFh, then after execution of instruction "SAL ax, 3", the result will be

- -3
- +3
- -8
- +8

Question No:30

If the decimal number "35" is shifted by two bit to left, the new value will be

- 35
- 70
- 140 (00100011 = 35 , 10001100=140)

➢ o17

Question No: 31

While using STOBS, if DF=1 then

➢ The value of SI will be incremented by one
- The value of SI will be incremented by two
- The value of SI will be decremented by one
- The value of SI will be decremented by two

Question No:32

After the execution of STOSW, the CX will be

- Decremented by 1
- **Decremented by 2 (Page 92)**
- Incremented by 1
- Incremented by 2

Question No: 33

The memory address always move from

- processor to memory
- memory to processor
- memory to peripheral

- peripheral to processor

Question No:34

An offset alone is not complete without

- **Segment (Page 34)**
- code label
- index register
- data label

Question No:35

Code Segment is associated to_____register by default.

- **IP (Page 34)**
- SS
- BP
- CX

Question No:36

After the execution of SAR instruction:

► MSB remain as it is

► MSB Will change

► MSB move to left (Page 52)

► No change will occur.

Question No:37

ASCII stands for _____.

The screen is two dimensional space having:

► 25 Rows and 25 Columns

► 25 Rows and 80 Columns

► 80 Rows and 80 Columns

► 80 Rows and 25 Columns (Page 80)

Question No:38

In the Rotate Right Operation every bit moves one position to right and the bit dropped from the right is inserted at the left and:

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► Dropped in CF (Page 53)

► moves to AL

► Don't go anywhere.

Question No:39

_____ of the following flags will be affected by MOVSB?

► DF

► ZF

► PF

► No effect on flags

Question No:40

The iAPX88 processor supports_____modes of memory access.

► 5

► 6

► 7 (Page 35)

► 8

Question No:41

Stack is a_____ that behaves in a first in last out manner.

► Program

► data structure (Page 67)

► Heap

► None of the Given

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Question No:42

The physical address of the stack is obtained by

► SS:SI combination

► SS:SP combination (Page 68)

► ES:BP combination

► ES:SP combination

Question No:43

Foreground and background parameter will be

► 32bits

► 16bits

► 8bits

► 4bits

Question No:44

The clear screen operation initialize whole block of memory

► 0741

► 0417

► 0714

► 0174, 0720 (Page 91)

Question No:45

In STOSB instruction, when DF is Set, SI is

► Incremented by 1

► Incremented by 2

► Decremented by 1

► Decremented by 2

Question No:46

Assembly language is:

► Low-level programming language

► High-level programming language

► Also known as machine language

► Not considered closer to the computer

Question No:47

A 32 Bit processor has accumulator of ---------

► 8 bit

► 16 bit

► 32 bit (Page 12)

► 64 bit

Question No:48

To transfer control back the RET instruction take

► 1 argument

► 1 argument

► 3 arguments (Page 72) (Not sure)

► No arguments

Question No:49

RET is executed, it recovers the values from

► Register

► Stack (Page 71)

► Data segment
► Code segment

Question No:50

To convert any digit to its ASCII representation

► Add 0x30 in the digit (Page 80)

► Subtract 0x30 from the digit

► Add 0x61 in the digit

► Subtract 0x61 from the digit

Question No:51

The prevalent convention in most high level languages is stack clearing by the

► Caller

► Callee (Page 74)

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► RET

► Stack

Question No:52

After execution of JCXZ instruction CX will changed with flag affect.

► CF

► OF

► DF

► None of Above (Page 43)

Question No:53

Execution of the instruction "mov word [ES : 0], 0x0741" will print

► "A" appear on the top left of screen (Page 81)

► "A" appear on the top right of screen

► "A" appear on the center of screen

► "A" appear on the bottom left of screen

Question No:54

if contains decimal -2 and BX contains decimal 2 then after the execution of instructions

CMP AX, BX JA label

► Jump will be taken

► Zero flag will set

► ZF will contain value -4

► Jump will not be taken

Question No:55

Which of the following options contain the set of instructions to open a window to the video memory?

► mov AX, 0xb008

mov ES, AX

► mov AX, 0xb800

mov ES, AX (Page 81) rep

► mov AX, 0x8b00

mov ES, AX

►mov AX, 0x800b

mov ES, AX

Question No:56

If D is “35” is shift to left 2 bits the new value

► 35

► 70

► 140

► 17

Question No:57

Execution of the instruction “mov word [ES : 0], 0x1230” will print the character color will

► Green

► White

► Red

► Black

Question No:58

After the execution of SAR instruction

► The msb is replaced by a 0

► The msb is replaced by 1

► The msb retains its original value

► The msb is replaced by the value of CF (Page 52)

Question No:59

RETF will pop the offset in the

► BP

► IP (Page 69)

► SP

► SI

Question No:60

Far” jump is not position relative but is _____

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► memory dependent

► Absolute (Page 46)

► Temporary

► indirect

Question No:61

After the execution of instruction “RET 2”

► SP is incremented by 2

► SP is decremented by 2

► SP is incremented by 4 (Page 66)

► SP is decremented by 4

Question No:62

DIV instruction has

► Two forms (Page 85)

► Three forms

► Four forms

► Five forms

Question No:63

When the operand of DIV instruction is of 16 bits then implied dividend will be of

► 8 bits (Page 85)

► 16 bits

► 32 bits

► 64 bits

Question No:64

After the execution of MOVS instruction which of the following registers are updated:

► SI only

► DI only

► SI and DI only (Page 92)

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► SI, DI and BP only

Question No:65

In the word designated for one screen location, the higher address contains

► The character code

► **The attribute byte (Page 81)**

► The parameters

► The dimensions

Question No:66

If the address of memory location Num1 is 0117 and its content is 0005 then after execution of the instruction " mov bx, Num1" bx will contain

►0005

►**0117 (Page 30)**

►Num1

►1701

Question No:67

In STOS instruction, the implied source will always be in

►**AL or AX registers (Page 92)**

►DL or DX registers

►BL or BX registers

►CL or CX registers

Question No:68

The shift logical right operation inserts

- A zero at right
- **A zero at left (Page 52)**
- A one at right
- A one at right

Question No:69

assembly the CX register is used normally as a_____register.

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►source

► counter (Page 32)

►index

►pointer

Question No:70

Which is the unidirectional bus ?

(I) Control Bus

(II) Data Bus

(III) Address Bus

►I only

►II only

►III only (Page 9)

►I and II only

Question No:71

JC and JNC test the_____flag.

►carry (Page 41)

►parity

►zero

►sign

Question No:72

After the execution of REP instruction CX will be decremented then which of the following flags will be affected?

►CF

►OF

►DF

► No flags will be affected (Page 93)

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Question No:73

which bit sets the character "blinking" on the screen?

►5

►6

►7 (Page 7)

►8

Question No:74

If we want to divide a signed number by 2, this operation can better be accomplished by

►SHR

► SAR (Page 52)

►SHL

►SAL

Question No:75

mov ax,5 has

►1 operand

► 2 opeand (Page 25)

►3 operand

►4 opeand

Question No:76

REP will always

► Increment CX by 1

► Increment CX by 2

► Decrement CX by 1 (Page 92)

► Decrement CX by 2

Question No:77

Index registers are used to store _____

►Data

►Intermediate result

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►Address (Page 16)

►Both data and addresses

Question No:78

The bits of the_____ work independently and individually

►index register

►base register

► flags register (Page 12)

►accumulator

Question No:79

Which mathematical operation is dominant during the execution of SCAS instruction

► Division

► Multiplication

► Addition

► Subtraction

Question No:80

The instruction ADC has_____Operand(s)

► 0

► 1

► 2

► 3 (Page 56)

Question No:81

In assembly the CX register is used normally as a_____register.

► source

► counter (Page 92)

► index

► pointer

Question No:82

When a 32 bit number is divided by a 16 bit number, the quotient will be stored in

► AX (Page 85)

► BX

► CX

► DX

Question No:83

In STOSB instruction SI is decremented or incremented by

►4

►1 (Page 92) SI and DI both are Index registers

►2

►3

Question No:84

CMPS instruction subtracts the source location to the destination location. Destination location always lies in

►DS:SI

►DS:DI

►ES:SI

► ES:DI (Page 93)

Question No:85

Regarding assembler, which statement is true:

► Assembler converts mnemonics to the corresponding OPCODE (Page 13)

► Assembler converts OPCODE to the corresponding mnemonics

► Assembler executes the assembly code all at once

► Assembler executes the assembly code step by step

Question No:86

If “BB” is the OPCODE of the instruction which states to “move a constant value to AX register”, the hexadecimal representation (Using little Endian notation) of the instruction “Mov AX,336” (“150” in hexadecimal number system) will be:

► 0xBB0150

► 0x5001BB

► 0x01BB50

► 0xBB5001 (Page 19)

Question No:87

one The maximum parameters a subroutine can receive (with the help of registers) are

► 6

► 7 (Page 72)

► 8

► 9

Question No:88

When a 16 bit number is divided by an 8 bit number, the dividend will be in

► AX (Page 85)

► BX

► CX

► DX

Question No:89

in Left-Shift-Operation the left most bit _____

► will drop

► will go into CF (Page 52)

► Will come to the right most

► will be always 1

Question No:90

Which of the following jumps makes intra segment control possible?

- Short
- **Far**
- Long
- Near

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Question No:91

There is an auto –increment mode when the ---------- is ----------.

- CX, not set
- ZF, not set
- DF, set
- **DF, not set**

Question No:92

REP allows the instruction to be --------CX times.

- Reported
- Replaced
- **Repeated**
- repositioned

Question No:93

In------- operation, a carry flag is inserted from the left moving every bit one position to its right. The right most bit is dropped into carry flag.

- **Rotate through carry (RCR)**
- Rotate Left (ROL)
- Rotate through carry left (RCL)
- Rotate Right(ROR)

Question No:94

ACSII stand for ---------- --.

- American Standard Code for information interaction
- **American standard code for information interchange**

Question No:95

---------- or ---------- are taken if it last arithmetic operation has produced a number in its destination that has odd parity.

- JP, JPE
- JS, JNS
- JCX, JNS
- **JNP, JPO**

Question No:96

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Which of the following string instruction is generally used in loop instead of REP prefix?

- SCAS
- **LODS**
- CMPS
- STOS

Question No:97

mov, ax [num 1] is an example of -------- bit move.

- 8
- **16**
- 24
- 32

Question No:98

Which of the following used data segment by default?

- Base pointer
- **Base register**
- Stack pointer
- Instruction pointer

Question No:99

XOR can also be used a ---------- to invert selective bits.

- Destination operand
- Source operand
- **Masking operator**
- Masking bit

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Question No:100

There are two ---------- from of MOVS instruction.

- 1
- **2**
- 3
- 4

Question No:101

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Registers are stored cells ----------.

- Outside the processor
- Both outside and inside the processor
- **Inside the processor**
- Inside the memory

Question No:102

When the operand of DIV instruction is in 16-bits, then the implied dividend will be in.

- AX
- **DX:AX**
- ES:AX
- DS:BX

Question No:103

Shifting the integer 5 left by 1 bit result in----

- 5
- **10**
- 25
- 50

Question No:104

In which of the following addressing both registers are not constant?

- Index +offset
- **Base+ Index**
- Base +offset
- Base+ offset+ index

Question No:105

---------- Transfers the word are top of stack (pointed to by Sp) to the destination operand and increment SP by two the new top of stack.

- PUSH
- **POP**
- CALL
- RET n

Question No:106

In------- operation, the output is 1 only if both input are 1.

- **AND**
- OR
- NOT
- XOR

Question No:107

Which of the following is used for calculating the segment address of an interrupt of an service routine “n”?

- **n*4+2**
- n*4-2
- n*4
- n*2+4

Question No:108

Conditional jump can only be --- --.

- Far
- **Short**
- Near
- long

Question No:109

Operand of POP is called----------since data is moving from the stack to the operand.

- source
- **destination**
- increment
- decrement

Question No:110

How many are the functions of a register?

- 3
- **2**
- 1
- 4

Question No:111

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SACS compares a source in accumulator register with the ----------string element addressed by ES:DI and updates the flags.

- Source
- **destination**
- flags
- register

Question No:112

which of the following instruction is used for non- destructive AND operation?

- CMP
- NAND
- **TEST**
- ADC

Question No:113

Simple CMP instruction uses --------- operation.

- Addition
- Division
- **Subtraction**
- Multiplication

Question No:114

What does the given instruction do?

SHR DL, 1

- **Move right most bit in carry**
- Move bit in DL
- Shift and store the One bit in DL
- Shift and store the most bit in Dl

Question No:115

--------- Subtracts one from the operand.

- MEC
- Modulus
- **DEC**
- Shift

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Question No:116

Which of following is an illegal assembly instruction?

- mov{num1}, al
- mov ax, bx
- mov al, bl
- **mov { num1},{num2}**

Question No:117

In assembly language, the first executable instruction of the program should be placed at the offset_____.

- 0*1100
- 0*0010
- 0*0000
- **0*0100**

Question No:118

Cs and Is are the both_____bit register.

- **16**
- 4
- 8
- 32

Question No:119

Which of the following is a special instruction?

- Add bx,0534
- **Cli**
- Mov ax,bx
- Cmp ax,0

Question No:120

Extended shifting algorithm consists of:

- 3 instructions
- 1 instructions
- **2 instructions**
- 4 instructions

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1. **Stack is a data structure that behaves in a First in Last …….. manner.**
   Out (Handsout searching Page 67)

2. **We can convert a digit into its ASCII representation by addingto it.**
   0x30 (Page 80)

3. **_____decrements SP (the stack pointer) by two and then transfers a word from the source operand to the top of stack?**
   Or_____function decrements SP (the stack pointer) by two and then transfers a word from the source operand to the top of stack now pointed to by SP.
   Push (page 71)

4. **The_____flag has a special role in debugging.**
   trap (Page 17)

5. **iAPX88 stands for:**
   Intel Advanced Processor Extensions 88 (Page 14)

6. **In_____every bit moves one position to the right and the bit dropped from the right is inserted at the left. This bit is alsocopied into the carry flag.**
   Rotate Right (ROR) (Page 53)

7. **Which of the following registers hold the address of data in**

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Intel Boss?



BX, BP, SI and DI **(page 30)**

8. **Which of the following is the general form of addressing?**
   Base + Index + Offset **(page 35)**

9. **All mathematical and logical operations are performed on the:**
   Accumulator **(page 12)**

10. **Which of the following register is used to hold the address of the next instruction to be executed?**
    Program Counter **(page 13)**

11. **Which of the following is a program control instruction?**
    jne 0534 **(page 14)**

12. **The basic function of register is to hold-------- --.**
    Oparands

13. **A 16-bit processor has an accumulator of:**
    16 bit **(Page 12)**

14. **REPE and REPNE prefixes are only meaningful with----------.**
    CMPS **(page 93)**

15. **Stack is a data structure that behaves in ----------manner.**
    LIFO
    Note: Stack two manner pr behave krta hy first FILO or second LIFO.Handsout mai LIFO Nhi hy magr iss ka correct answer LIFO hy.

16. **Which of the following bit is dropped into the carry flag after the execution of "Shift Logical Right" operation?**
    Right most bit **(page 52)**

17. **How logical errors are different from syntax error?**

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Identifying syntax and logical errors it the responsibility of assembler and programmer respectively.

18. **In___operation, the output is 1 only if both inputs are 1.**
    AND

19. **How jmp is different form jnz?**
    jmp is an unconditional jump, however jnz is a conditional jump

20. **In Shift Logical Left (SHL), the_____bit is dropped into the carry flag.**
    most significant (page 52)

21. **In STOS, the implied source operand always resides in------- --.**
    AL or AX (page 92)

22. **Which of the following is an illegal assembly instruction?**
    mov [num1], [num2] (page 30)

23. **sub sp, 2**
    The above instruction is used to perform a_____operation on stack.
    PUSH (page 75, 76)

24. **The maximum amount of memory accessible using 8085 processor is_____.**
    64 KB (page 15)

25. **In direct addressing, the memory address given in theinstruction is___:**
    fixed (page 28)

26. **What makes CMP different from SUB?**

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SUB is a destructive subtraction while CMP is a non-destructive subtraction. Both affect the flags.

27. Which of the following are the basic bitwise logical operations?
AND, OR, XOR, and NOT (page 58)

28. To convert the case of character, we add or subtract_____from it its ASCII code
0x20 (page 80)

29. When the RET instruction is executed, it recovers the value of the_____from the stack.
instruction pointer (page 68)

30. When the relative address stored with the instruction is in 16 bits, the jump is called a_____jump.
Near (page 45)

31. Registers are storage cells_____.
inside the processor (page 11)

32. _____subtracts one from the operand.
DEC (page 55)

33. STI stands for?
Set the Interrupt flag

34. RET instruction takes_____arguments(s).
0 (page 64)

35. A symbol associated to a point in the program is called a_____.
Or A symbol associated to an instruction in the program is known as:
Label (page 25)

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36. In 8088, there is a_____stack.
decrementing (page 68)

37. CX register is mostly used as a_____register.
Counter (page 32)

38. There are_____byte(s) for each character on the screen.
2 (page 87)

39. Whenever an instruction needs a memory source, holds the pointer to it.
DS:SI (page 91)

40. Keeping in view the downward compatible between the two systems, the codes written for the intel iAPX88 are_____on the intel 386 processor.
valid (page 14)

41. In multiplication algorithm, We take the first digit of the multiplier and multiply it with the_____.
multiplicand (page 51)

42. The ordering of a program's instructions is ensured by the ---- register.
Program Counter (page 13)

43. When an element is pushed on the stack, SP is decremented by ……..
two (page 68)

44. In iAPX88,_____flag is specifically related to string instructions.
direction (page 17)

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45. Which of following conditional jump is not dependant on any flag?

Or Which of the following jump does not depend on flags register?

JCXZ (page 43)

46. In --------- operation, a zero is inserted from the left and every bit moves one position to the right. The right most bit is dropped or down into the carry flag.

Shift Logical Right (SHR) (page 52)

47. In case of DIV BX instruction, the quotient is stored in ------- register.

AX (page 85)

48. Which of the following flags sets when a large number is subtracted from a smaller number?

CF (page 88)

49. Which of the following is the most illegal instruction?

Mov al, ax

50. In ADC instruction, there are --------- --.

two operands and the CF (page 57)

51. Call instruction takes a ---------as an argument.

Label (page 64)

Or Call instruction takes -------- argument(s).

1 (page 64)

52. VGA stands for_____.

Video Graphics Adapter (page 79)

53. When the control is transferred to a subroutine?

After the arguments are pushed on the stack

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54. CS and IP are both --------- bit registers.
16 (page 21)

55. A decrementing stack moves from -------- to ---------- addresses as elements are added in it.
higher, lower (page 68)

56. The most convenient place to store local variables is--------- --.
stack (page 75)

57. can be used to check whether particular bits of anumber are set or not.
AND (page 60)

58. Which of the following operations is used to set any specific bit in a binary number?
OR (page 59)

b. Which of the following operations is used to clear any specific bit in a binary number?
AND (page 59)

59. Which bit of attribute byte represents the blue component of foreground color?
0 (page 81)

60. The segment offset pair is called a/an ---------- address.
logical (page 22)

61. The extra bit produced as a result of an arithmetic operation that does not fit in the target register is stored in:
carry flag (page 16)

62. Cell width refers to the total number of bits in a memory cell while the total number of cells is called the ----------.

depth (page 10)

63. Intel 8085 can access up to ---------of memory, whereas Intel 8088 can access up to ---------- of memory.
64 KB, 1 MB (page 15)

64. Registers are normally part of----------.
CPU

65. The base pointer accesses local variables using --------- offsets.
negative (page 75)

66. ASCII table is the contiguous arrangement of the uppercase alphabets (41-5A), the lowercase alphabets (61-7A), and the numbers -------- --.
30-39 (page 80)

67. Which of the following string instructions is generally used in a loop instead of REP prefix?
LODS (page 92)

68. The correct instruction to subtract with borrow is ---------.
SBB (page 57)

69. mov [1234], ax is an example of ---------- addressing.
direct (page 35)

70. What the following piece of code does?
Shl word [multiplicand], 1
Rcl word [multiplicand+2], 1
Extended Shift left (page 58)

71. When the execution of Call instruction, the value of----------is decremented by 2.
SP (page 66)

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72. **Which of the following bus is used to inform the memory that whether processor wants to read data or write data?**
Control Bus **(page 9)**

73. **In far jump,---------- --.**
both offset and segment are given **(page 46)**

74. **A value 0500 is stored in memory. If we transfer this value to a general-purpose register, then it will be shown as:**
0005 **(page 28)**

75. **In which of the following addressing, both registers are not constant?**
Base + Index **(page 36)**

76. **The number of---------- in a cell is called the cell width.**
Bits **(page 10)**

77. **In extended multiplication, we store the multiplicand in ------ bits, and the result is stored in ------- bits.**
32, 32 **(page 57)**

78. **Synchronization between the processor and the memory for the read and write operations is done by the --------- --.**
Control bus **(page 11)**

79. **iAPX88 Architecture consists of:**
14 Registers **(page 15)**

80. **Shift logical Left (SHL) moves all the bits one position to the --- and inserts a zero from the ------ --.**
left, right **(page 52)**

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81. In MUL instruction, if the source operand is a byte, then it is multiplied with ---------- register.
AL (page 87)

b. In MUL instruction, if the source operand is a word, then it is multiplied with ---------- register.
AX (page 87)

82. When the operand of DIV instruction is in 16 bits, then the implied dividend will be in --------- --.
DX:AX (page 85)

83. What does the PUSH operation do?
its copies the operand on the stack (page 68)

84. ---------- transfers the word at the top of stack (pointed to by SP) to the destination operand and then increments SP by two.
POP (page 71)

85. Which of the following is an incorrect XOR operation?
XOR BX,AX (page 59)

86. This jump is taken if the last arithmetic operation changed the sign unexpectedly.
JO (page 43)

87. Which of the following jumps is taken if the last arithmetic operation did not changed the sign unexpectedly?
JNO (page 43)

88. Which would be the output of MOV AX, BL?
Assembler will declare that this is an illegal instruction (page 30)

89. ------- or ------- are taken if the last arithmetic operation has produced a number in the destination that has add parity.

JNP, JNO (page 43)

90. Group of bits processor uses to inform memory which element to read/write is collectively known as:
Address bus (page 9)

91. MOVS is used to move a block of memory.
Move (page 92)

92. Stack Pointer is ---------- when the ret instruction is executed.
incremented by 2 (page 68)

93. instruction directs the flow of program.
Program Control (page 14)

94. Which of the following instructions allows memory to memory movement of data?
MOVS (page 91)

95. By default, IP is associated with----------.
CS (page 34)

96. After the execution of Shift Arithmetic Right instruction, the most significant bit --------- --.
retains its original value (page 52)

97. In case of DIV BX instruction, the quotient is stored in ------ register.
AX (page 85)

98. Which part of B80500 (encoded instruction) is an opcode?
B8 (pager 25)

99. The operand of ------- is called the source operand because the data moves to the stack from the operand.

PUSH **(page 71)**

100. **Which of the following jump is taken if the last arithmetic operation has produced a number in its destination that has even parity?**
JPE **(page 43)**

101. **When there is a 16 bit operand with the DIV instruction, then the implied dividend will be in ----------bits.**
32 **(page 85)**

102. **EBCDIC, grey code and ASCII are the standards for --------- representation of characters.**
alpha-numeric

103. **In ---------, a zero is inserted from the right and every bit moves one position to its left with the most significant bit dropped into the carry flag.**
both SHL and SAL **(page 52)**

104. **Conditional jumps can only be --------- --.**
Short **(page 46)**

105. **BP stands for---------- pointer.**
Base **(page 16)**

106. **In video memory, each screen location corresponds to ---- bytes.**
2 **(page 82)**

107. **What makes CMP different from SUB?**
SUB is a destructive subtraction while CMP is a non-destructive subtraction. Both affect the flags.

108. **Which of the following registers is used as a counter?**

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CX **(page 15)**

109. **The memory on video controller is accessible to the processor by ---------bus.**
System **(page 80)**

110. **Which of the following addressing scheme has been used in the instruction MOV [BX], AX?**
Based Register Indirect **(page 35)**

111. **The multiplier is stored in ---- bits in Extended Multiplication.**
16 **(page 57)**

112. **Which part of machine code tells the central processor to perform a certain task?**
Operation code

113. **Which of the following elements will come out first using the POP instruction if 6,3,5,8,4,7 and 9 were pushed on the stack? 9**

114. **During program execution, if any change occurs in AH or AL,is also reflected in ----------.**
AX **(page 15)**

115. **To modify the processor's behavior, ---------- instructions are used.**
special **(page 14)**

116. **Which of the following uses data segment by default?**
Base register or BX **(page 35)**

117. **The operand of POP is called the --------- because the data is moving from the stack to the operand.**
destination **(page 71)**

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118. In assembly language, the first executable instruction of the program should be placed at the offset -------- --.
0x0100 (page 26)

119. Suppose, the current value of IP is 0x0129, and the relative address is 0x0012. What will be the new value of IP?
0x013B

120. In the instruction “mov word [es:0], 0x1230”, 12 refers to ------ color on--------- background.
green, blue (page 81)

121. mov ax, [num1] is an example of ---------- bit move.
8

122. The physical memory address of 1234:5678 segment-offset pair is:
179B8

123. XOR can also be used as a ---------to invert selective bits.
masking operator (page 60)

124. Call instruction fchanges the values of ----- and ------- registers.
IP, SP (page 65)

125. There are ------- registers in iAPX88 architecture that can hold address of data.
4 (page 80)

126. What does the given instruction do?
SHR DL, 1
Move the right most bit into the carry flag (page 55)

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127. **In which of the following terms, a character is stored in memory while writing assembly language program?**
ASCII code **(page 82)**

128. **Physical memory address is of:**
20 bits **(page 21)**

129. **In 8088, there is a----------stack.**
decrementing **(page 68)**

130. **Which of the following is also called intro-segment cell?**
Near call **(page 72)**

131. **Suppose BX=0x0120, CS=0x1000, and the memory under consideration is [CS:BX+0x0880].**
Which of the following will be the physical address?
109A0

132. **Which of the following is the correct syntax for 'OR' operationin assembly language?**
or ax, bx **(page 59)**

133. **In assembly language, "DIV BL" instruction has an---------- bit operand.**
8 **(page 85)**

134. **Flags register is a special register in every architecture. It is also known as --------.**
program status word **(page 12)**

135. **An important role of the stack is in the creation of ----- variables that are only needed while the subroutine is in execution and not afterwards.**
Local **(page 75)**

136. **Stack is:**

a data structure (page 67)

137. Constant can never be used as ---------.
destination (page 25)

138. SP is associated (by default) with --------- --.
SS (page 34)

139. Which bit of attribute byte represents the green component of foreground color?
1 (page 81)

140. BP stands for---------- pointer.
Base (page 32)

141. Shift logical Right (SHR) moves all the bits one position to the --------- and inserts a zero from the------- --.
right, left (page 52)

142. The execution of the instruction "mov word [ES :160], 0x1230" will print a character on the screen at the -------- column of the ---------- row.
first, second (page 81)

143. The stack of iAPX88 works on -------- sized elements.
word (page 68)

144. MOVS is used to---------- a block of memory.
Move (page 92)

145. Which bit of attribute byte represents the red component of foreground color?
2 (page 81)

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146. ---------- can also be used as a masking operation to invert selective bits.
XOR (page 60)

147. What is the content of stack pointer?
Address of the top element of the stack

148. Far jump is not position relative rather---------- --.
Absolute (page 46)

149. In MUL instruction, if the source operand is a byte then the result is retuned in ------- and------ --.
AH, AL (page 87)

150. Which of the following are the two variants of STOS instruction?
STOSB and STOSW (page 92)

151. ---------- cannot appear as an operand in any instruction, it is used for accessing instructions.
IP

152. Which of the following instructions allows code reusability in 8088?
CALL (page 64)

153. Which bit of the attribute byte represents the red component of foreground color?
2 (page 81)

154. In 8051, there is an---------- stack.
incrementing (page 68)

155. Shift logical Right (SHR) operation copies the ------ bit into the carry flag.
right most (page 52)

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156. **Shift Arithmetic Right (SAR) shifts every bit one place to the right and places a copy of the ------- significant bit at the ----- significant place.**
most, most (page 52)

157. **There are --------columns per row on the screen.**
80 (page 87)

158. **In XOR operation, the output if 1 if both inputs are-------- --.**
different (page 59)

159. **Which of the following is the extension of object file?**
.exe

160. **The stack pointer marks the ---------- of stack.**
top (page 68)

161. **Shifting the integer 5 left by 1 bit results in------- --.**
10

162. **Program consists of ------- logical parts.**
Two (page 20)

163. **In -------- operation, a carry flag is inserted from the left moving every bit one position to its right. The right most bit is dropped in the carry flag.**
Rotate Through Carry Right (RCR) (page 53)

164. **Memory address always goes from ……**
processor to memory

165. **The unit of stack operations in iAPX88 is a ……**
word (page 68)

166. 8088 is a ……. processor with its accumulator and all registers of ……..
16 bit, 16 bits (page 14)

167. Which of the following keywords is used to define two bytes inmemory?
DW (page 25)

168 ……jump is absolute and not position relative.
Far (page 46)

169. Which of the following directive is used to reserve an eight bitsspace in memory?
db (page 25)

170. What makes db different from dw?
dw occupies 16 bits and db occupies 8 bits (page 25)

171. The total numbers of symbols in extended ASCII are …….
256

172 ……. is used for permanent diversion.
Jump (page 63)

173. In MUL instruction, if the source operand is a word, then it is multiplied with ----------register.
AX (page 87)

174. The intent of the processor is carried to the memory for read/write operations through the……
control bus (page 11)